

bounded disturbance input.

Chapter 6 Input-to-State Stability for stochastic Switched Systems.

6.1 Problem Formulation

$$\begin{cases} dx(t) = f_{\sigma(t)}(t, x(t), u(t))dt + g_{\sigma(t)}(t, x(t), u(t))dw(t) & (6.1a) \\ x(t_0) = x_0 & (6.1b) \end{cases}$$

$$x(t_0) = x_0 \quad (6.1b)$$

$x \in \mathbb{R}^n$ — state vector (right-continuous stochastic process)

$u: [t_0, \infty) \rightarrow \mathbb{R}^l$ — input ($\|u(t)\|_{\infty} \leq 1$)

$\sigma(t): [t_0, \infty) \rightarrow \mathcal{J}$ — switching signal ($\mathcal{J} = \{1, 2, \dots, N\}$)

Markovian switching: $T = [\tau_{ij}]_{N \times N}$ $\tau_{ij} \geq 0, i \neq j, \tau_{ii} = -\sum_{j=1, j \neq i}^N \tau_{ij}$

$$P(\sigma(t+h) = j | \sigma(t) = i) = \begin{cases} \tau_{ij}h + o(h), & i \neq j \\ 1 + \tau_{ii}h + o(h), & i = j \end{cases}$$

$\{t_k\}_{k \in \mathbb{N}}$: switching times. $t_k \uparrow$ and $\lim_{k \rightarrow \infty} t_k = \infty$.

$f_i: [t_{k-1}, t_k) \times \mathbb{R}^n \times \mathbb{R}^l \rightarrow \mathbb{R}^n, g_i: [t_{k-1}, t_k) \times \mathbb{R}^n \times \mathbb{R}^l \rightarrow \mathbb{R}^{n \times m}$

$f_i(t, 0, 0) = 0, g_i(t, 0, 0) = 0 \Rightarrow x \equiv 0$ is a trivial solution.

x_0 — initial state ($\mathcal{F}_0$ -measurable, $E[\|x_0\|^2] < \infty$)

$f_i, g_i \in \mathcal{L}_{ad}(\Omega, L^p([t_{k-1}, t_k]))$
to guarantee a unique solution.

Lemma 6.1

Assume $x(t) = x(t; t_0, x_0)$ is the solution to

$$dx(t) = f(t, x(t), u(t))dt + g(t, x(t), u(t))dw(t), x(t_0) = x_0.$$

radially bounded.

τ_{∞} — the explosion time. $V \in C^{1,2}(\mathbb{R}_+ \times \mathbb{R}^n; \mathbb{R}_+)$, $\lim_{\|x\| \rightarrow \infty} [\inf_{t \geq t_0} V(t, x)] = \infty$.

$$LV(t, x, u) \leq \alpha(V(t, x)), \text{ (a.s.) } \alpha \in \mathcal{K}_c.$$

Then $E(V(t, x(t))) \leq G^{-1}[G(E(V(t_0, x_0))) + (-t_0)] < \infty \quad \forall t \geq t_0$

where $G(s) = \int_1^s \frac{dt}{\alpha(t)}$. G^{-1} — inverse function of G .

Moreover, $x(t)$ is unique and defined for all $t \geq t_0$.

Proof: ① Claim $\tau_{\infty} = \infty$ (By contradiction)

If not true, $\exists \varepsilon > 0, T > 0$ s.t. $P\{\tau_{\infty} \leq T\} > \varepsilon$.

$$\tau_l \leq \tau_{\infty}$$

Define stopping times $\tau_l (l \geq 1)$: $\tau_l = \inf\{t \geq t_0 \mid \|x(t)\| > l\} \Rightarrow \lim_{l \rightarrow \infty} \tau_l = \tau_{\infty}$ (a.s.)

$\Rightarrow P\{\tau_l \leq T\} \geq P\{\tau_{\infty} \leq T\} > \varepsilon$

For all $t \in [t_0, T]$, let $\tau_l(t) = \min\{\tau_l, t\} \Rightarrow \tau_l(t) = \begin{cases} \tau_l, & \tau_l \leq t \leq T \\ t, & t_0 \leq t \leq \tau_l \end{cases}$

By Itô formula $\Rightarrow$

i) For $t \leq \tau_1$, $\tau_1(t) = t$, $\forall s \in [t_0, t]$, $\tau_1(s) = s \Rightarrow \int_{t_0}^{\tau_1(t)} V(s, x(s), u(s)) ds = \int_{t_0}^t V(\tau_1(s), x(\tau_1(s)), u(\tau_1(s))) ds \leq \int_{t_0}^t \alpha(V(\tau_1(s), x(\tau_1(s)), u(\tau_1(s)))) ds$

ii) For $\tau_1 \leq t \leq T$, $\tau_1(t) = \tau_1 \leq t$, $\forall s \in [t_0, \tau_1]$, $\tau_1(s) = s \Rightarrow \int_{t_0}^{\tau_1(t)} V(s, x(s), u(s)) ds = \int_{t_0}^{\tau_1} V(\tau_1(s), x(\tau_1(s)), u(\tau_1(s))) ds \leq \int_{t_0}^{\tau_1} \alpha(V(\tau_1(s), x(\tau_1(s)), u(\tau_1(s)))) ds \leq \int_{t_0}^{\tau_1} \alpha(V(\tau_1(s), x(\tau_1(s)), u(\tau_1(s)))) ds$

$$\begin{aligned} E(V(\tau_1(t)), x(\tau_1(t))) &= E(V(t_0, x_0)) + E \int_{t_0}^{\tau_1(t)} V(s, x(s), u(s)) ds \\ (*) &\leq E(V(t_0, x_0)) + E \int_{t_0}^t \alpha(V(\tau_1(s), x(\tau_1(s)))) ds \\ &\leq E(V(t_0, x_0)) + \int_{t_0}^t \alpha(E(V(\tau_1(s), x(\tau_1(s)))) ds \quad \alpha \text{ is concave} \end{aligned}$$

$$\begin{aligned} \text{By Bihari's inequality} \Rightarrow E[V(\tau_1(t), x(\tau_1(t)))] &\leq G^{-1}[G(E[V(t_0, x_0)]) + (t - t_0)] \\ &\leq G^{-1}[G(E[V(t_0, x_0)]) + (T - t_0)] \end{aligned}$$

On the other hand,

$$E(V(\tau_1(T), x(\tau_1(T)))) = E(V(\tau_1(T), x(\tau_1(T))) \mathbb{1}_{(\tau_1 \leq T)}) + E(\underbrace{V(\tau_1(T), x(\tau_1(T)))}_{\geq 0} \underbrace{\mathbb{1}_{(\tau_1 > T)}}_{\geq 0})$$

$$\begin{aligned} \text{Then } E(V(\tau_1(T), x(\tau_1(T))) \mathbb{1}_{(\tau_1 \leq T)}) &\leq G^{-1}[G(E[V(t_0, x_0)]) + (T - t_0)] \\ &= E(V(\tau_1, x(\tau_1)) \mathbb{1}_{(\tau_1 \leq T)}) \end{aligned}$$

Define $\eta_l = \inf \{V(t, x) \mid \|x\| \geq l, t \geq t_0\}$. $\|x(\tau_1)\| \geq l$ by the definition of τ_1

$$\Rightarrow G^{-1}[G(E[V(t_0, x_0)]) + (T - t_0)] \geq E(V(\tau_1, x(\tau_1)) \mathbb{1}_{(\tau_1 \leq T)}) \geq \eta_l P(\tau_1 \leq T) > \eta_l$$

Letting $l \rightarrow \infty$ since $\lim_{\|x\| \rightarrow \infty} \inf_{t \geq t_0} V(t, x) = \infty \Rightarrow \eta_l \rightarrow \infty$

$\Rightarrow$ Contradiction $\Rightarrow \tau_{\infty} = \infty$

② Prove $E[V(t, x(t))] \leq G^{-1}[G(E[V(t_0, x_0)]) + t - t_0]$

$$\begin{aligned} E[V(t, x(t))] &\leq E[V(t_0, x_0)] + E \int_{t_0}^t V(s, x(s), u(s)) ds \\ &\leq E[V(t_0, x_0)] + E \int_{t_0}^t \alpha(V(s, x(s))) ds \\ &\leq E[V(t_0, x_0)] + \int_{t_0}^t \alpha(E(V(s, x(s)))) ds \end{aligned}$$

By Bihari's inequality to get the result.

③ local Lipschitz condition $\Rightarrow$ uniqueness.

Def 6.2.

i) uniformly asymptotically ISS (aISS) in the p th moment if $\exists \beta \in \mathcal{KL}$ and $\gamma \in \mathcal{K}$ s.t. $\forall u$, and $p > 1$.

$$E[\|x(t)\|^p] \leq \beta(E[\|x_0\|^p], t - t_0) + \gamma(\|u(t)\|_{\infty}), \quad \forall t > t_0 \quad (t_0 \in \mathbb{R}_+)$$

where $E[\|x(t)\|^p] < \infty$, $x(t) = x(t; t_0, x_0)$ — solution of (6.1).

ii) exponentially ISS (eISS) in the p th moment if, in addition

$$\beta(E[\|x_0\|^p], t - t_0) \leq K E[\|x_0\|^p] e^{-\lambda(t - t_0)} \text{ for some } K > 0, \lambda > 0.$$

$\mathcal{K} \triangleq \{\alpha \in C([0, \infty); \mathbb{R}_+); \uparrow, \alpha(0) = 0\}$, $\mathcal{K}_{\infty} \triangleq \{\alpha \in C([0, \infty); \mathbb{R}_+); \uparrow, \alpha(0) = 0, \alpha(r) \rightarrow \infty \text{ as } r \rightarrow \infty\}$

$\mathcal{KL} \triangleq \{\beta \in C([0, \infty) \times \mathbb{R}_+; \mathbb{R}_+); \beta(\cdot, s) \in \mathcal{K}, \beta(r, \cdot) \searrow, \beta(r, s) \rightarrow 0 \text{ as } s \rightarrow \infty\}$.

6.2 Initial-State-Dependent Dwell-time

Switching at fixed times $\{t_k\}$ ↙ all stable modes

Theorem 6.1 (aISS under the initial-state dependent on dwell-time condition)

$p \geq 1$. For $i \in \mathcal{I}$, $V_i \in C^{1,2}([t_{k-1}, t_k] \times \mathbb{R}^n; \mathbb{R}_+)$ with $V_i(t, 0) = 0, \forall t \in [t_{k-1}, t_k]$.

(i) $\exists \alpha_{1i} \in \mathcal{K}_\infty$, concave, $\alpha_{2i} \in \mathcal{K}_\infty$, convex, s.t.

$$\alpha_{2i}(\|x\|^p) \leq V_i(t, x) \leq \alpha_{1i}(\|x\|^p) \quad \text{a.s.} \Rightarrow \|x\|^p \geq \alpha_{1i}^{-1}(V_i(t, x))$$

(ii) $\exists \alpha_{3i} \in \mathcal{K}_V, \gamma \in \mathcal{K}$, s.t.

$$\mathcal{L}V_i(t, x(t), u(t)) \leq -\alpha_{3i}(\|x\|^p) \quad \text{a.s.}$$

provided that $\|x\|^p > [\alpha_{3i}^*]^{-1}(\frac{1}{\gamma} \gamma(\|u\|_\infty)) =: \rho_i(\|u\|_\infty)$ (a.s.), where $0 < \nu < 1$, $\alpha_{3i}^*(\cdot) = \frac{1}{1-\nu} \alpha_{3i}(\cdot)$.

(iii) $\forall k \in \mathbb{N}$, τ_{isd} condition holds: $\tau_{isd} \geq \ln \frac{\theta_{2i}(a_{k-1} E[\|x\|^p])}{\theta_{1i}(a_k E[\|x\|^p])}$

$a_k \downarrow_0, a_0 = 1, \theta_{1i}, \theta_{2i} \in \mathcal{K}_\infty$.

$\Rightarrow$ (6.1) is p th moment aISS with ISS-gain

$$\rho_M(\cdot) = \max\{\rho_i(\cdot) = \alpha_{3i}^*]^{-1}(\gamma^*(\cdot)) \mid i \in \mathcal{I}\}, \quad \gamma^* = \frac{1}{\gamma} \gamma.$$

Proof: ① Define $m_i(t) = E(V_i(t, x(t))) \quad t \in [t_{k-1}, t_k]$.

Itô formula + taking expectation + Fubini Theorem + $\begin{cases} \alpha_{1i} \text{ concave} \\ \alpha_{2i} \text{ convex} \end{cases}$

$$\begin{aligned} \Rightarrow m_i(t) &= m_i(t_{k-1}) + E \int_{t_{k-1}}^t \mathcal{L}V_i(s, x(s), u(s)) ds \\ &\leq m_i(t_{k-1}) - E \int_{t_{k-1}}^t \alpha_{3i}(\|x(s)\|^p) ds \\ &\leq m_i(t_{k-1}) - E \int_{t_{k-1}}^t \alpha_{3i}(\alpha_{1i}^{-1}(V_i(s, x(s)))) ds \\ &= m_i(t_{k-1}) - \int_{t_{k-1}}^t E[\alpha_{3i}(V_i(s, x(s)))] ds \\ &\leq m_i(t_{k-1}) - \int_{t_{k-1}}^t \alpha_{3i}(m_i(s)) ds \end{aligned}$$

By Bihari's inequality, $G_i(t) = \int_t^t \frac{1}{\alpha_{3i}(s)} ds$

$$m_i(t) \leq G_i^{-1}[G_i(m_i(t_{k-1})) + (t - t_{k-1})] \triangleq \beta_i^*(m_i(t_{k-1}), t - t_{k-1}), \quad \beta_i^* \in \mathcal{KL}.$$

PUP By [4.5], $\exists \theta_{1i}^*, \theta_{2i}^* \in \mathcal{K}_\infty$ s.t. $\beta_i^*(s, t) \leq \theta_{1i}^*(\theta_{2i}^*(s) \exp(-t))$.

$$\alpha_{2i}(E[\|x(t)\|^p]) \leq m_i(t) \leq \theta_{1i}^*(\theta_{2i}^*(m_i(t_{k-1})) e^{-(t-t_{k-1})}) \leq \theta_{1i}^*(\theta_{2i}^*(\alpha_{1i}(E[\|x(t_{k-1})\|^p])) e^{-(t-t_{k-1})})$$

$$\Rightarrow E[\|x(t)\|^p] \leq \alpha_{2i}^{-1} \theta_{1i}^*(\theta_{2i}^*(\alpha_{1i}(E[\|x(t_{k-1})\|^p])) e^{-(t-t_{k-1})}) \triangleq \theta_{1i}^{-1}(\theta_{2i}(E[\|x(t_{k-1})\|^p]) e^{-(t-t_{k-1})})$$

where $\theta_{1i}(\cdot) = (\theta_{1i}^*)^{-1}(\alpha_{2i}(\cdot))$, $\theta_{2i}(\cdot) = \theta_{2i}^*(\alpha_{1i}(\cdot))$

② For $t \in [t_0, t_1]$, $E[\|x(t)\|^p] \leq \theta_{1i}^{-1}[\theta_{2i}(E[\|x(t_0)\|^p]) e^{-(t-t_0)}]$

first mode $\rightarrow$ By τ_{isd} condition, at switching time $t = t_1$,

$$E[\|x(t_1)\|^p] \leq \theta_{1i}^{-1}[\theta_{2i}(E[\|x(t_0)\|^p]) e^{-(t_1-t_0)}] \leq \alpha_{1i} E[\|x_0\|^p]$$

$$\stackrel{\tau_{isd}}{\Rightarrow} \theta_{2i}(E[\|x_0\|^p]) \leq \theta_{1i}(\alpha_{1i} E[\|x_0\|^p]) \Rightarrow \leq \theta_{1i}^{-1}[\theta_{1i}(\alpha_{1i} E[\|x_0\|^p])] \rightarrow$$

second mode
↓

$$\theta_{22}(a_1 \in [\|x_0\|_P]) \leq \theta_{12}(a_2 \in [\|x_0\|_P]) e^{\tau_{isd}}$$

$$\Rightarrow \theta_{12}^{-1}(\theta_{22}(a_1 \in [\|x_0\|_P]) e^{\tau_{isd}}) \leq a_2 \in [\|x_0\|_P]$$

For $t \in [t_1, t_2]$, $E[\|x(t_2)\|_P] \leq \theta_{12}^{-1}[\theta_{22}(E[\|x(t_1)\|_P]) e^{-(t_2-t_1)}]$

τ_{isd} condition $\hookrightarrow \leq \theta_{12}^{-1}[\theta_{22}(a_1 \in [\|x_0\|_P]) e^{-(t_2-t_1)}] \leq a_2 \in [\|x_0\|_P]$

For $t \in [t_{k-1}, t_k]$, $E[\|x(t_k)\|_P] \leq \theta_{11}^{-1}[\theta_{21}(a_{k-1} \in [\|x_0\|_P]) e^{-(t-t_{k-1})}] \Rightarrow E[\|x(t_k)\|_P] \leq a_k \in [\|x_0\|_P]$

whenever $\|x(t_k)\| > [\rho_i(\|u\|_\infty)]^{1/p}$

Since $\lim_{k \rightarrow \infty} a_k = 0$, $\exists \tau, s.t. \frac{\tau}{\|x(t_k)\|} = [\rho(\|u\|_\infty)]^{1/p}$

Theorem 6.2. (stable modes and unstable modes)

$\mathcal{I} = \mathcal{I}_s \cup \mathcal{I}_u$, $V_i \in C^{1,2}([t_{k-1}, t_k] \times \mathbb{R}^n; \mathbb{R}_+)$, $V_i(t, 0) = 0$.

(i) $\forall i \in \mathcal{I}$, $\exists \alpha_{1i} \in \mathcal{K}_\infty$, concave, $\alpha_{2i} \in \mathcal{K}_\infty$, convex, s.t.

$\alpha_{2i}(\|x\|_P) \leq V_i(t, x) \leq \alpha_{1i}(\|x\|_P)$. (a.s.)

(ii) (1) $\forall i \in \mathcal{I}_s$, $\exists \alpha_{3i} \in \mathcal{K}_U$, $\rho_i \in \mathcal{K}_\infty$, s.t.

$\dot{V}_i(t, x, u) \leq -\alpha_{3i}(\|x\|_P)$, (a.s.), whenever $\|x\|_P > \rho_i(\|u\|_\infty)$.

(2) $\forall i \in \mathcal{I}_u$, $\exists \alpha_{3i} \in \mathcal{K}_C$, s.t.

$\dot{V}_i(t, x, u) \leq \alpha_{3i}(\|x\|_P)$. (a.s.)

(iii) τ_{isd} condition :

(1) $\forall i \in \mathcal{I}_s$, $k=1, 3, 5, \dots$

$t_1 - t_0 \geq \ln \frac{\theta_{21}(E[\|x_0\|_P])}{\theta_{11}(a_1 \in [\|x_0\|_P])} > 0$

$t_3 - t_2 \geq \ln \frac{\theta_{23}(a_1, A_1 \in [\|x_0\|_P])}{\theta_{13}(a_2 \in [\|x_0\|_P])} > 0$

...

(2) $\forall i \in \mathcal{I}_u$, $k=2, 4, 6, \dots$

$0 < t_2 - t_1 \leq G_{12}[\alpha_{22}(a_1, A_1 \in [\|x_0\|_P])] - G_{12}[\alpha_{12}(a_1 \in [\|x_0\|_P])]$

$0 < t_4 - t_3 \leq G_{14}[\alpha_{24}(a_2, A_2 \in [\|x_0\|_P])] - G_{14}[\alpha_{14}(a_2 \in [\|x_0\|_P])]$

...

where $0 < a_k < a_k A_k \leq a_{k-1}$ with $a_0 = 1$. $\theta_{1i}(\cdot) = \theta_{1i}^*(\alpha_{1i}(\cdot))$ and $\theta_{2i}(\cdot) = \theta_{2i}^*(\cdot) \in \mathcal{K}_\infty$. $G_2, G_4, \dots$ defined in Lemma 6.1.

$\Rightarrow$ (b.1) is the pth moment aISS with the ISS gain $\rho_M(\cdot) \triangleq \max_{1 \leq i \leq p} \rho_i(\cdot)$.

6.3. Markovian Switching.

Markovian switching to control the mode switching.

$$(6.7) \begin{cases} dx(t) = f(t, x(t), u(t), \sigma(t)) dt + g(t, x(t), u(t), \sigma(t)) dW(t). \\ x(t_0) = x_0, \quad \sigma(t_0) = \sigma_0 \in J. \end{cases}$$

$$V(t, x(t), i) \in C^{1,2}(\mathbb{R}_+ \times \mathbb{R}^n \times J; \mathbb{R}_+), \quad i \in J.$$

$$\begin{aligned} \mathbb{E} V(t, x(t), u(t), i) &= V_t(t, x(t), i) + V_x(t, x(t), i) f(t, x(t), u(t), i) + \frac{1}{2} \text{tr} [g^T(t, x(t), u(t), i) V_{xx}(t, x(t), i) g(t, x(t), u(t), i)] \\ &\quad + \sum_{j=1}^N \gamma_{ij} V(t, x(t), j). \end{aligned}$$

Theorem 6.3 (eISS in pth moment)

For any $i \in J$, $\forall t \in [t_{k-1}, t_k]$.

$$(i) \exists k > 0, \alpha_i > 0, \rho_i \geq 0, \sigma_i \geq 0 \text{ s.t.} \quad (a.s.)$$

$$\|f(t, x, 0, i)\| \leq k \|x\|, \quad \|x^T f(t, x, 0, i)\| \leq \alpha_i \|x\|^2.$$

$$\|g(t, x, 0, i)\| \leq \rho_i \|x\|, \quad \|x^T g(t, x, 0, i)\| \leq \sigma_i \|x\|^2. \quad (a.s.)$$

$$(ii) \exists \lambda > 0, C_1 > 0, C_2 > 0 \text{ s.t.}$$

$$C_1 \|x\|^p \leq V(t, x, i) \leq C_2 \|x\|^p. \quad (a.s.)$$

$$\mathbb{E} V(t, x, u, i) \leq -\lambda \|x\|^p. \quad (a.s.)$$

Whenever $\|x\| > \rho(\|u\|_\infty)$, $\rho \in \mathbb{R}$, $V \in C^{1,2}([t_0, \infty) \times \mathbb{R}^n \times J; \mathbb{R}_+)$.

$$(iii) f, g \text{ Locally Lipschitz in } u, \text{ uniformly in } t, x, \text{ i.e., } \exists C_3, C_4 > 0 \text{ s.t.}$$

$$\|f(t, x, u, i) - f(t, x, 0, i)\| \leq C_3 \|u\|, \quad (a.s.)$$

$$\|g(t, x, u, i) - g(t, x, 0, i)\| \leq C_4 \|u\|, \quad (a.s.)$$

$$\Rightarrow (6.7) \text{ is pth moment eISS for } 0 < p < \min \{2, (\exists C_4 + 4\sigma_i) / (C_4 + 2\sigma_i)\}.$$

$$\text{Lyapunov exponent} \leq -\lambda / C_2.$$

Proof: $i \in J$, $\beta_i > 0$, define $V(t, x(t), i) = \beta_i \|x(t)\|^p$ — Lyapunov function candidate of i th mode.

$$\begin{aligned} \Rightarrow \mathbb{E} V(t, x, u, i) &= p \beta_i \|x\|^{p-2} x^T f(t, x, u, i) + \frac{1}{2} p \beta_i \|x\|^{p-2} \|g(t, x, u, i)\|^2 - \frac{1}{2} p(p-2) \beta_i \|x\|^{p-4} \|x^T g(t, x, u, i)\|^2 \\ &\quad + \sum_{j=1}^N \gamma_{ij} \beta_j \|x\|^p. \end{aligned}$$

$$\leq -\beta_i^* \|x\|^p + 2M(\|x\|) \|u\|_\infty$$

$$\leq -\lambda^* \|x\|^p + 2M(\|x\|) \|u\|_\infty.$$

$$\text{where } \beta_i^* = -[\beta_i(p|\alpha_i| + \frac{1}{2}p\rho_i^2 + \frac{1}{2}p(2-p)\sigma_i^2) + \sum_{j=1}^N \gamma_{ij}\beta_j] > 0$$

$$M(\|x\|) = \max \{ \beta_i [C_3 + p\rho_i C_4] \|x\|^{p-1}, C_4 p \beta_i [-(0.5C_4 + \sigma_i)p + (1.5C_4 + 2\sigma_i)] \|x\|^{p-2} \}.$$

To use $\lambda^* \|x\|^p$ to dominate $2M(\|x\|) \|u\|_\infty$,

$$\begin{aligned} L(t, x, u, i) &\leq -(\lambda^* - \nu) \|x\|^p - \nu \|x\|^p + 2M(\|x\|) \|u\|_\infty \quad 0 < \nu < \lambda^* \\ &\leq -(\lambda^* - \nu) \|x\|^p = -\lambda \|x\|^p \end{aligned}$$

provided that $\nu \|x\|^p > 2M(\|x\|) \|u\|_\infty$ or

$$\|x\| > 2\beta_i / \nu \cdot [C_3 + p\rho_i C_4] \|u\|_\infty \quad \text{if } M(\|x\|) = \beta_i [C_3 + p\rho_i C_4] \|x\|^{p-1}$$

$$\|x\| > \{2\beta_i p C_4 / \nu \cdot [-(0.5C_4 + \sigma_i)p + (1.5C_4 + 2\sigma_i)] \|u\|_\infty\}^{\frac{1}{p-2}}$$

$$\text{if } M(\|x\|) = C_4 p \beta_i [-(0.5C_4 + \sigma_i) + (1.5C_4 + 2\sigma_i)] \|x\|^{p-2}$$

(i)
 $\Rightarrow$ exponential decay